

Multi-pathway exposure assessment: EXCAM Project for Infants in Rural Ethiopia

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Ref: Wang Y, Yang Y, Slanzi CM, Li X, Ojeda A, Paro F, et al. (2025) Quantitative multi-pathway assessment of exposure to *Escherichia coli* for infants in Rural Ethiopia. PLoS Negl Trop Dis 19(6): e0013154.

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- ① Study objective and design
- ② Data collection: behavioral observations + environmental sampling
- ③ Model architecture
 - Duration-based behavior: competing hazards (state transitions)
 - Frequency-based behavior: inhomogeneous Poisson point process
 - Contamination: left-censored lognormal
 - Multi-pathway exposure ABM: 7 modules (sources–vehicles–sinks)
- ④ Simulation workflow (10,000 child-days)
- ⑤ Key results (behavior, contamination, dominant pathways)

Objective

Goal

Quantify infants' exposure to fecal contamination (indicator: *E. coli*) through multiple pathways in rural Ethiopia, and compare exposure profiles across two infant age windows.

Key questions

- Which pathways dominate ingestion dose at early vs late infancy?
- How do behavior sequences (where/when/how) shift pathway importance?
- How do environmental contamination levels propagate through hand/food vehicles into ingestion?

Study Design: EXCAM (Rural Ethiopia)

Population and timepoints

- Study site: Haramaya woreda, rural Ethiopia
- Infants enrolled: $n = 79$ at T1; $n = 76$ at T2 (3 dropouts)
- T1: 4–8 months; T2: 11–15 months

Data streams

- ① Structured behavioral observations (~1300 hours)
- ② Human + environmental microbiological samples ($N = 1338$)

Paired design (household-level)

- Same households observed at two developmental stages
- Behavior observation → informs sampling locations next day

Exposure metric

Daily ingestion dose (in CFU/day) by source and pathway.

Behavioral Observations: What is Recorded?

Two classes of behaviors

- **Duration-based:** awake, sleeping, bathing, drinking (recorded as start–end intervals)
- **Frequency-based:** touching, mouthing, pica, eating (recorded as event timestamps)

Context variables

- **Compartment:** carried by mother / carried by other / on surface with barriers / on bare ground
- **Location:** within homestead / out of homestead

State definition (for duration layer)

$$S(t) = (\text{activity}, \text{compartment}, \text{location})$$

Environmental and Human Sampling (E. coli)

Sample types (interfaces)

Areola swab; breast milk; mother handrinse; sibling handrinse; infant handrinse; bathing water; drinking water; fomites; food; soil.

Measurement and units

- Lab methods return MPN or CFU; assume $1 \text{ MPN} \equiv 1 \text{ CFU}$
- Use $\log_{10}$ concentration for modeling (with detection limit)

Linking design

Sampling locations are selected based on enumerators' observation notes (behavior-informed household sampling).

Duration-based Behavior Model: Competing Hazards

State transitions

Let S_k denote the current state. A transition occurs when the child changes activity/compartment/location.

Competing risks / hazards

From state i , multiple next states $j \in \mathcal{J}(i)$ compete. Define a transition-specific hazard:

$$\lambda_{ij}(t) \geq 0, \quad j \in \mathcal{J}(i)$$

Total hazard out of state i :

$$\lambda_i(t) = \sum_{j \in \mathcal{J}(i)} \lambda_{ij}(t)$$

Duration-based Behavior Model: Competing Hazards

Implications

- Draw next event time (waiting time) via $Exp(\lambda_i)$
- Select destination state j proportional to $\lambda_{ij}(t)$ at that time
- Fit transition rates in a Bayesian framework (JAGS), then simulate full-day sequences

Frequency-based Behavior Model: Inhomogeneous Poisson

State-conditional event intensity

For behavior b (e.g., touching fomites), intensity depends on current state s :

$N_b(\cdot)$ is a Poisson point process with $\lambda_b(t) = \lambda_{b,S(t)}$

Rate estimation

Let $N_{b,s}$ be total events of type b observed during times the child is in state s ; let T_s be total time in s .

$$\hat{\lambda}_{b,s} = \frac{N_{b,s}}{T_s}$$

Simulation (time-rescaling idea)

Rescale time in each state by $\hat{\lambda}_{b,s}$ to simulate events via a homogeneous Poisson process and map back to original timeline.

Contamination Model: Left-censored Lognormal

Model

For each sample type m and timepoint,

$$Y = \log_{10}(C) \sim \mathcal{N}(\mu_m, \sigma_m^2)$$

with left-censoring at $\log_{10}(\text{LLOD})$.

Contamination Model: Left-censored Lognormal

Likelihood contribution

For an observed positive sample ($C \geq \text{LLOD}$):

$$\ell_{\text{pos}}(\mu, \sigma; y) = \log \left(\frac{1}{\sigma} \phi \left(\frac{y - \mu}{\sigma} \right) \right)$$

For a censored (negative) sample ($C < \text{LLOD}$):

$$\ell_{\text{cens}}(\mu, \sigma) = \log \left(\Phi \left(\frac{\log_{10}(\text{LLOD}) - \mu}{\sigma} \right) \right)$$

Estimation

Estimate (μ_m, σ_m) by MLE separately for each sample type and timepoint.

Exposure ABM: Sources, Vehicles, Sinks

Network abstraction

- **Sources:** soil, food, breast milk, areola surface, drinking water, fomites, hands of mother/other adults/children
- **Vehicles:** hands and food (carry microbes between nodes)
- **Sinks:** mouth (ingestion); bathing water (detachment/ingestion mechanism)

State variables tracked (conceptually)

$\mathbf{N}_H(t) \in \mathbb{R}_+^K$ (microbes on hands by original source)

$\mathbf{N}_I(t) \in \mathbb{R}_+^K$ (ingested microbes by original source)

where K indexes original sources.

Exposure ABM: 7 Transfer Modules

Modules (behavior-triggered)

- 1 Hand touching: attachment/detachment via touching hands/fomites
- 2 Mouthing: ingestion via mouthing hands (caregiver/infant) and fomites
- 3 Bathing: bathing water ingestion + detachment from hands
- 4 Pica: soil ingestion + hand-soil contact
- 5 Eating: ingestion of solid food (+ possible hand-food contact)
- 6 Breastfeeding: breast milk ingestion + areola mouthing
- 7 Drinking water: ingestion of drinking

Generic transfer calculation

When an event occurs (touch/mouth/eat/pica), compute transfer from node A to B :

$$\Delta N_{A \rightarrow B} = f_{\text{transfer}}(\text{contact type, surface, dose, } \theta)$$

Update bookkeeping vectors:

$$\mathbf{N}_H \leftarrow \mathbf{N}_H + \Delta \mathbf{N}_{\rightarrow H}, \quad \mathbf{N}_I \leftarrow \mathbf{N}_I + \Delta \mathbf{N}_{\rightarrow I}$$

Simulation Workflow (per timepoint)

- 1 Simulate a “typical child day” (14-hour daytime) duration-based state sequence via competing hazards
- 2 Conditional on state timeline, simulate frequency-based event streams via inhomogeneous Poisson
- 3 Sample contamination levels for each interface from fitted left-censored lognormal models
- 4 Run ABM transfer modules along the day to accumulate ingestion by source
- 5 Repeat for 10,000 simulated child-days per timepoint

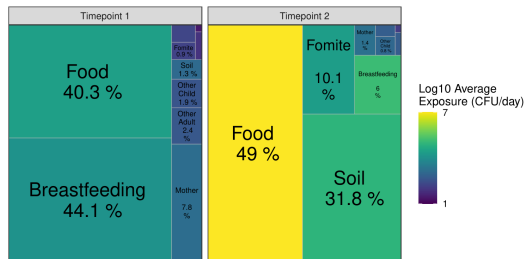
Main output

$$\text{Dose}_k = \sum_{t \in [0, 14h]} \Delta N_{I,k}(t) \quad (\text{CFU/day})$$

for each source k .

Results: Dominant Exposure Pathways

- Infant exposure to fecal contamination was simulated over 10,000 typical child-days.
- Dominant exposure pathways differ by age.
- **Timepoint 1 (4–8 months)**
 - Food
 - Breastfeeding
- **Timepoint 2 (11–15 months)**
 - Food
 - Soil

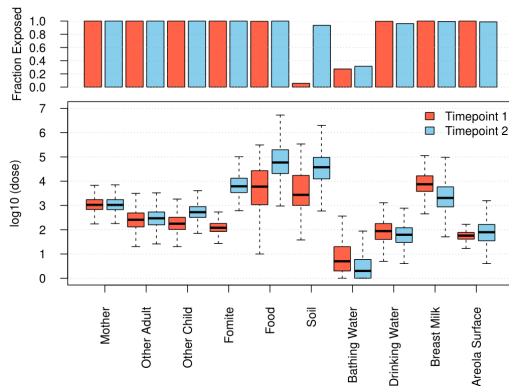


Exposure Dose Distribution

- Daily ingestion dose varies substantially across pathways.
- Food, soil and breastfeeding generate the largest exposure doses.
- Exposure is generally higher at Timepoint 2.

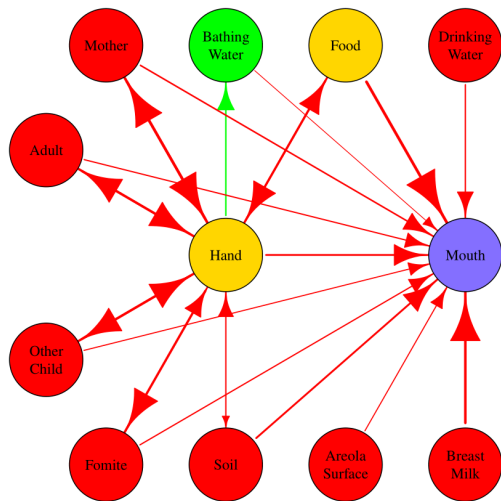
Interpretation

Behavioral development increases exposure through environmental pathways.



Fecal Microbe Transfer Network(eg. TP1)

- Microbes move between environmental sources, hands and ingestion.
- Direct ingestion pathways dominate exposure:
 - Eating
 - Breastfeeding
 - Soil ingestion (pica)
 - Mouthing fomites
- Hand-mediated pathways contribute less to total exposure.



Key Findings of the Study

- Infant exposure to fecal contamination occurs through multiple interconnected pathways.
- Dominant pathways shift with infant development.
 - Timepoint 1: food and breastfeeding
 - Timepoint 2: food and soil
- Behavioral changes strongly influence exposure risks.
- Results highlight the importance of:
 - hygienic breastfeeding practices
 - safe food handling
 - prevention of soil ingestion

From *E. coli* to *Campylobacter*: What Changes?

Components that remain the same

- behavior model
- state transitions
- event simulation
- exposure pathways
- agent-based transfer structure

Components that must be updated

- source-specific contamination inputs
- transfer assumptions by pathway
- pathogen survival / persistence assumptions
- exposure metric and dose scale

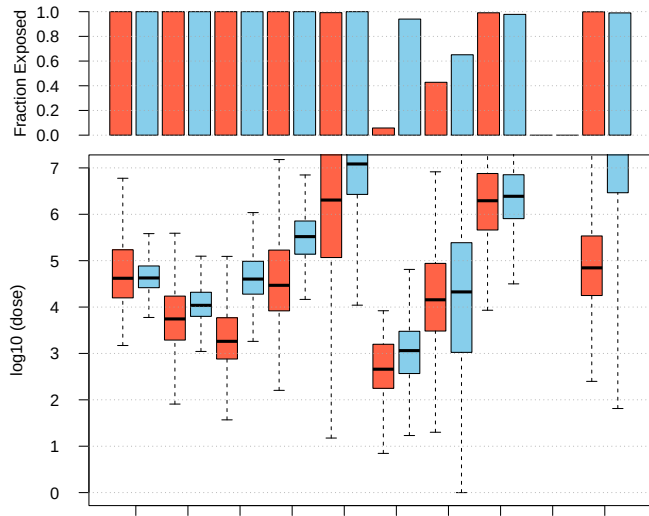
Core modeling idea

The EXCAM architecture can be retained, while the contamination layer and pathogen-specific parameters are replaced.

Preliminary Results from the *Campylobacter* Extension

- The full simulation workflow has been executed under the migrated model.
- Preliminary outputs include:
 - pathway-specific exposure contribution,
 - source-wise exposure summaries,
 - simulated daily dose patterns.

Preliminary Results from the *Campylobacter* Extension



Preliminary Results from the *Campylobacter* Extension

	Timepoint 1 (May–Dec 2021)	Timepoint 2 (Dec 2021–June 2022)
# subjects	158	152
mean age (range) in days	184 (132–263)	371 (338–466)
duration of observation (hours)	698	608
# duration-based activity transitions	3044	2517
average % of time spent awake	63.6	77.1
average % of time spent bathing	0.2	0.2
average % of time spent drinking	14.3	10.5
average % of time spent sleeping	21.9	12.1
# frequency-based activities	23677	30801
rate of eating (times per hour)	0.36	2.20
rate of mouthing (times per hour)	13.27	10.68

Preliminary Results from the *Campylobacter* Extension

	Timepoint 1 (May–Dec 2021)	Timepoint 2 (Dec 2021–June 2022)
# compartment transitions	2753	3780
average % of time spent carried by mother	40.3	26.0
average % of time spent carried by other	14.5	6.9
average % of time spent on the bare ground	0.2	16.6
average % of time spent on a surface with barriers	45.0	50.5
# location transitions	44	150
average % of time spent out of homestead	0.4	2.2
average % of time spent within homestead	99.6	97.8

Preliminary Results from the *Campylobacter* Extension

Table: Number of samples and censoring proportion by environmental source.

Source code	Source	Total samples	Censored samples	Proportion censored
EB	Bathing water	44	14	0.32
ED	Drinking water	148	76	0.51
EF	Food	96	87	0.91
ES	Soil	148	48	0.32
EV	Fomite	265	234	0.88
HA	Areola swab	151	150	0.99
HB	Breast milk	143	143	1.00
HH	Sibling hand rinse	125	69	0.55
HP	Mother hand rinse	152	94	0.62
HR	Infant hand rinse	152	121	0.80

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Source code	Source	Parameter	Mean	SD	2.5%	50%	97.5%	n_{eff}
EB	Bathing water	μ	3.68	0.23	3.22	3.68	4.10	2312
		σ	1.39	0.20	1.05	1.37	1.86	2576
ED	Drinking water	μ	3.11	0.13	2.84	3.11	3.35	1415
		σ	1.21	0.12	1.01	1.20	1.47	1379
EF	Food	μ	-0.51	1.58	-4.35	-0.19	1.25	297
		σ	2.62	1.01	1.48	2.41	5.23	289
ES	Soil	μ	3.97	0.08	3.81	3.98	4.13	1945
		σ	0.87	0.07	0.75	0.87	1.01	2343
EV	Fomite	μ	0.65	0.50	-0.45	0.71	1.47	827
		σ	2.15	0.34	1.60	2.11	2.94	801
HA	Areola swab	μ	3.00	0.15	2.69	3.01	3.28	1631
		σ	1.30	0.14	1.05	1.29	1.60	1655
HH	Sibling hand rinse	μ	2.74	0.17	2.36	2.75	3.04	1440
		σ	1.42	0.16	1.15	1.40	1.77	1296
HP	Mother hand rinse	μ	1.95	0.31	1.20	1.99	2.44	718
		σ	1.45	0.24	1.07	1.42	2.01	717
HR	Infant hand rinse	μ	2.74	0.17	2.36	2.75	3.04	1440
		σ	1.42	0.16	1.15	1.40	1.77	1296